Figure 1

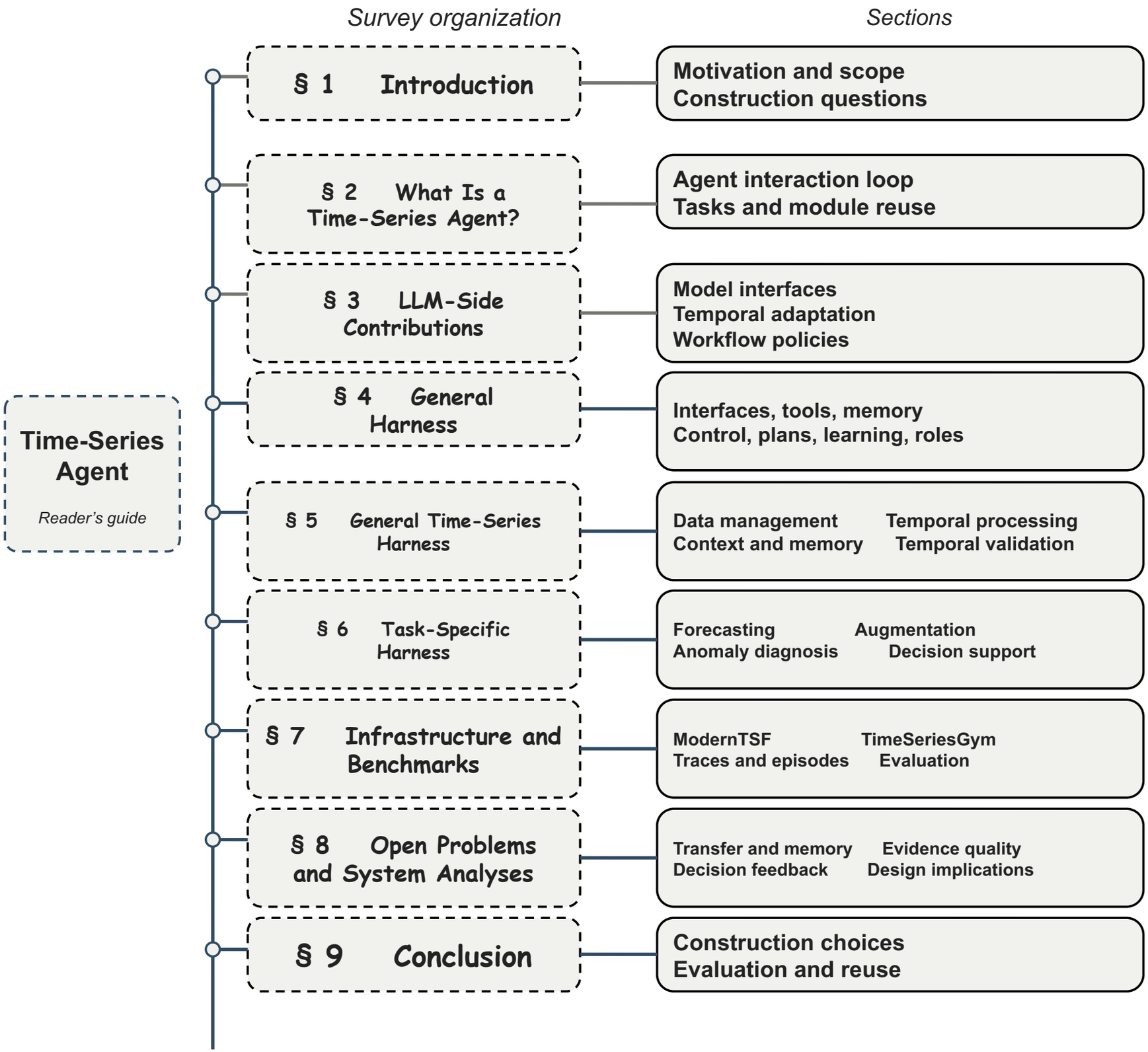

Survey organization
Sections
Time-Series Agent
Reader's guide
§ 1 Introduction
Motivation and scope
Construction questions
§ 2 What Is a Time-Series Agent?
Agent interaction loop
Tasks and module reuse
§ 3 LLM-Side Contributions
Model interfaces
Temporal adaptation
Workflow policies
§ 4 General Harness
Interfaces, tools, memory
Control, plans, learning, roles
§ 5 General Time-Series Harness
Data management
Temporal processing
Context and memory
Temporal validation
§ 6 Task-Specific Harness
Forecasting
Augmentation
Anomaly diagnosis
Decision support
§ 7 Infrastructure and Benchmarks
ModernTSF
TimeSeriesGym
Traces and episodes
Evaluation
§ 8 Open Problems and System Analyses
Transfer and memory
Evidence quality
Decision feedback
Design implications
§ 9 Conclusion
Construction choices
Evaluation and reuse

Figure 2

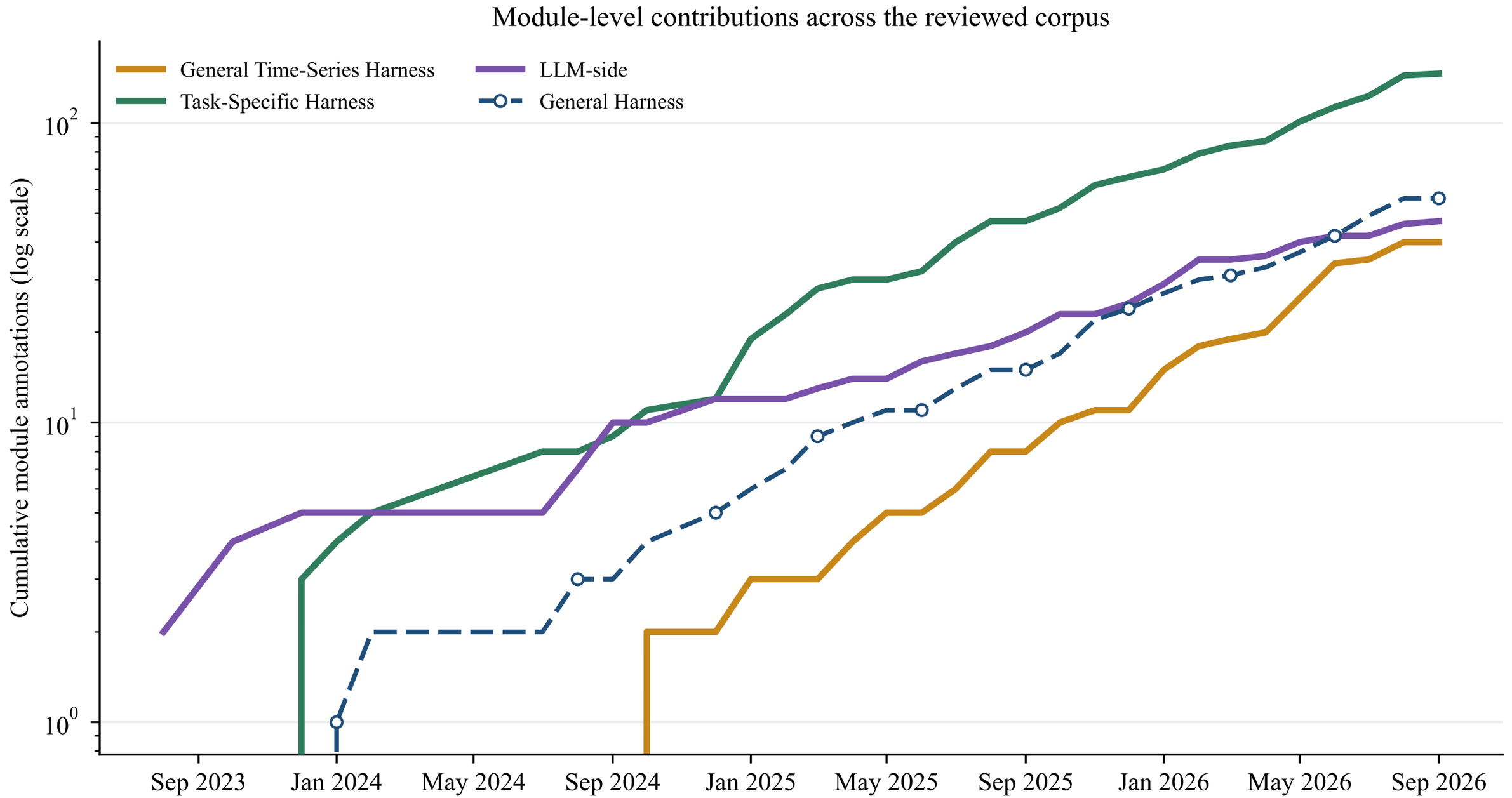

Module-level contributions across the reviewed corpus
General Time-Series Harness
LLM-side
Task-Specific Harness
General Harness
Cumulative module annotations (log scale)
10^2
10^1
10^0
Sep 2023
Jan 2024
May 2024
Sep 2024
Jan 2025
May 2025
Sep 2025
Jan 2026
May 2026
Sep 2026

Figure 3

Model-Side Contributions

Representation · Alignment
Understanding · Adaptation

Representation & Alignment



- Time-LLM
- LLM4TS
- LLMTime
- ChatTS

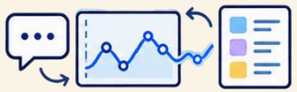
Learning & Reasoning



- Time-Series Reasoning
- STRIDE
- OpenForecaster
- AURA-4B

Harness Mechanisms for Time-Series Agent Systems

1 Interface & Interaction



Structured Analysis Interfaces

- TS-Agent
- TimeART
- TSAG
- TESSA

Adaptive Interaction & Observation

- SAGE-Sleep
- ARTIST
- TimeRLM
- LLM-TSFD

2 Memory & Context



Retrieval & Context Construction

- Agentic RAG for TSA
- TimeCAP
- From News to Forecast
- ZARA
- MERIT

Temporal Experience & Evidence Memory

- FinMem
- MemCast
- ForecastCompass
- TIEM

3 Tools & Execution



Model & Code Execution

- TimeCopilot
- TimeSeriesScientist
- MoiraiAgent
- GenAutoML

Domain Tools & Simulation

- AgriWorld
- Open-TI
- GridMind
- PPG Physiological Agent

4 Control, Verification & Feedback



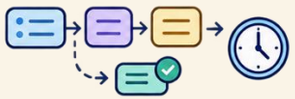
Review & Adjudication

- Argos
- AnomaMind
- TSQAgent
- Oil-Well Explainable Agent

Validation & Acceptance Gates

- Grid-Agent
- HydroAgent
- Neuro-Agentic Control
- BatteryLake

5 Planning & Workflow



Workflow Construction

- ClimateAgent
- DCATS
- MOSAIC
- Nexus

Operational Analysis Pipelines

- HFMD Structured Agents
- SIREN
- Smart Energy Agent
- S2S-FDD

6 Optimization & Evolution



Reflection & Experience Updates

- TimeClaw-EEL
- FLAIRR-TS
- CastFSR
- CRAFTER

Search & Adaptive Optimization

- MetaCaster
- EvoTS-Agent
- MarsTSC
- ATLAS

7 Coordination & Scaling



Debate & Evidence Aggregation

- TS-Debate
- MAS4TS
- Competition Agents
- FactorMAD

Coordinated Decision Systems

- FinCon
- TradingAgents
- CoLLMLight
- Fin-Analyst

Evaluation & Infrastructure

Benchmarks, datasets, simulators and task protocols for time-series agents

Benchmarks & Evaluation Resources



- TemporalBench
- TFRBench
- TraceBench
- TSAQA
- MMTS-Bench
- TimeSeriesGym
- TSQBench
- TimeSage-MT
- TS-Skill
- TimeSeriesExamAgent
- FutureSim
- WorldReasoner

Infrastructure & Task Protocols



- AION
- ModernTSF
- TimeClaw-Harness

Figure 4

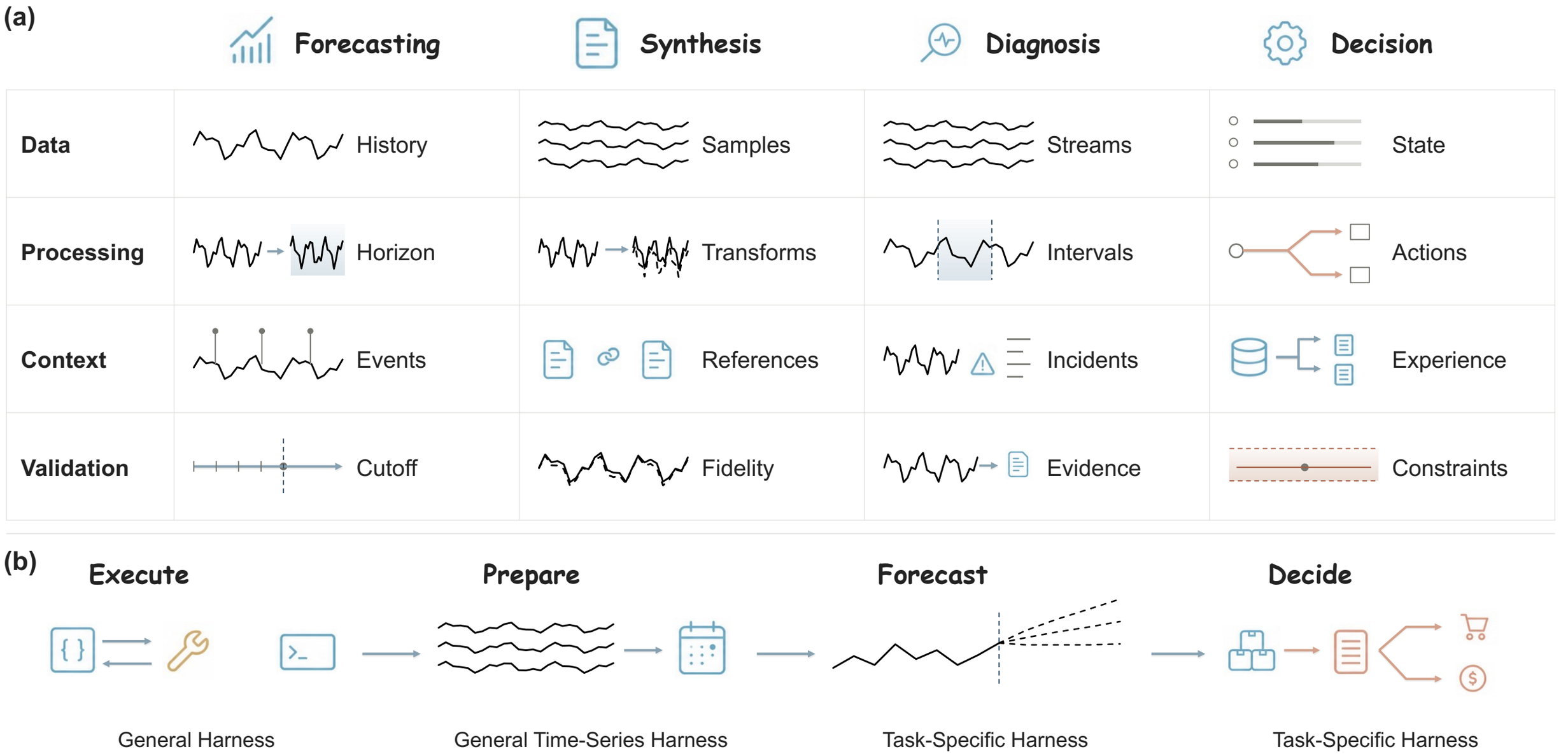

(a)
Forecasting
Synthesis
Diagnosis
Decision
Data
History
Samples
Streams
State
Processing
Horizon
Transforms
Intervals
Actions
Context
Events
References
Incidents
Experience
Validation
Cutoff
Fidelity
Evidence
Constraints
(b)
Execute
Prepare
Forecast
Decide
General Harness
General Time-Series Harness
Task-Specific Harness
Task-Specific Harness

Figure 5

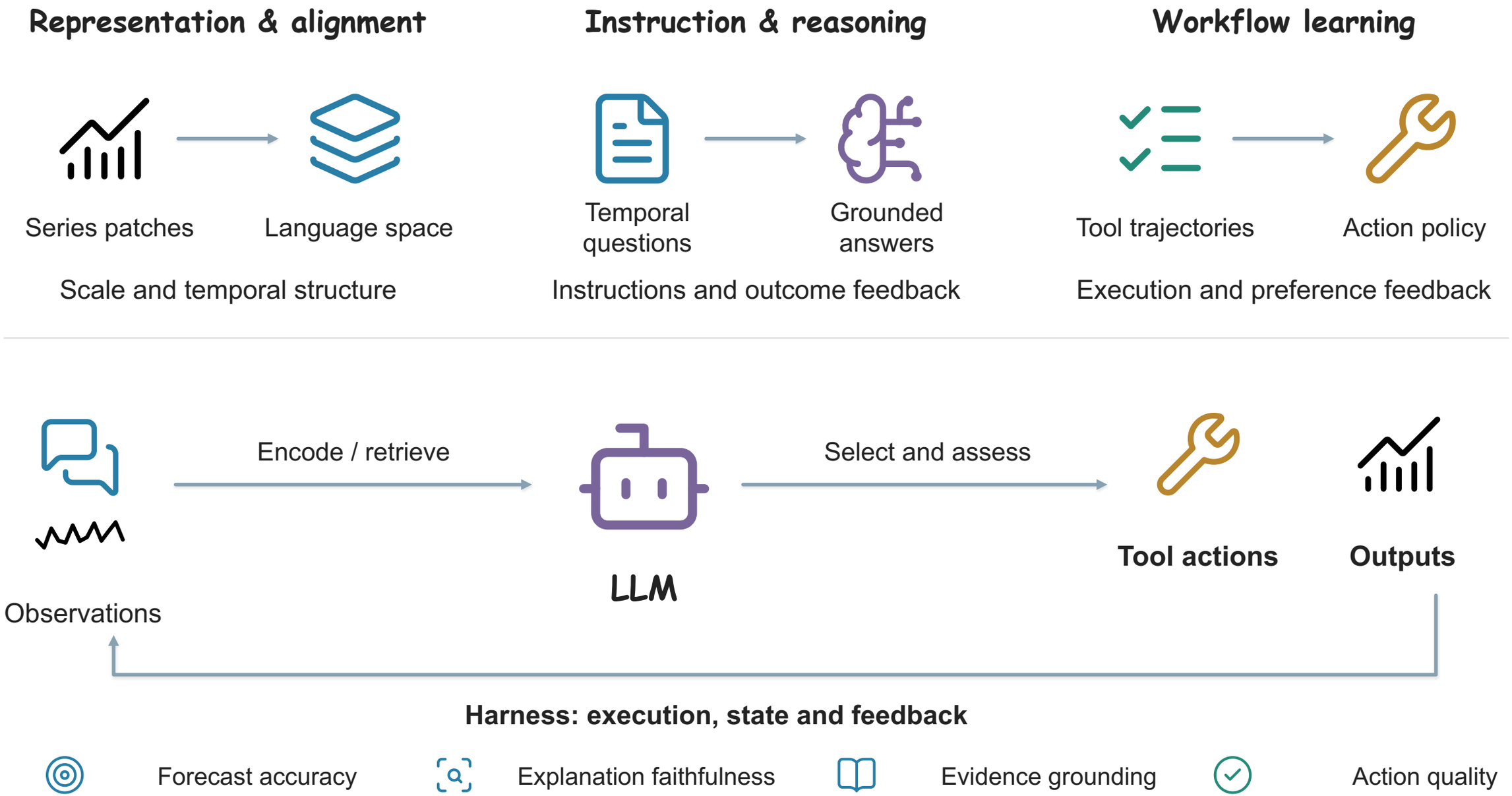

Representation & alignment
Instruction & reasoning
Workflow learning
Series patches
Language space
Temporal questions
Grounded answers
Tool trajectories
Action policy
Scale and temporal structure
Instructions and outcome feedback
Execution and preference feedback
Encode / retrieve
Select and assess
Observations
LLM
Tool actions
Outputs
Harness: execution, state and feedback
Forecast accuracy
Explanation faithfulness
Evidence grounding
Action quality

Figure 6

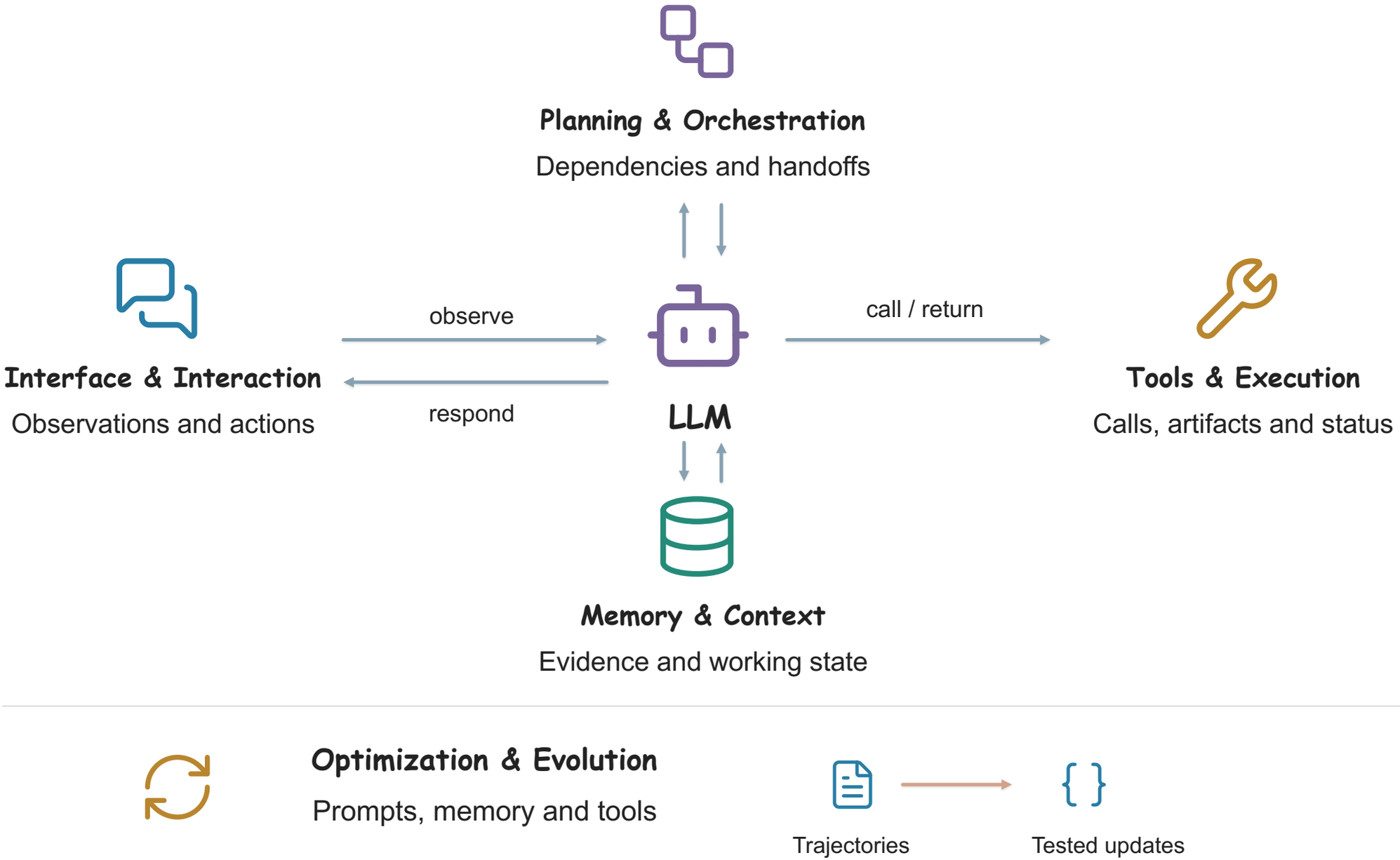

Planning & Orchestration
Dependencies and handoffs
Interface & Interaction
Observations and actions
observe
respond
LLM
call / return
Tools & Execution
Calls, artifacts and status
Memory & Context
Evidence and working state
Optimization & Evolution
Prompts, memory and tools
Trajectories
Tested updates

Figure 7

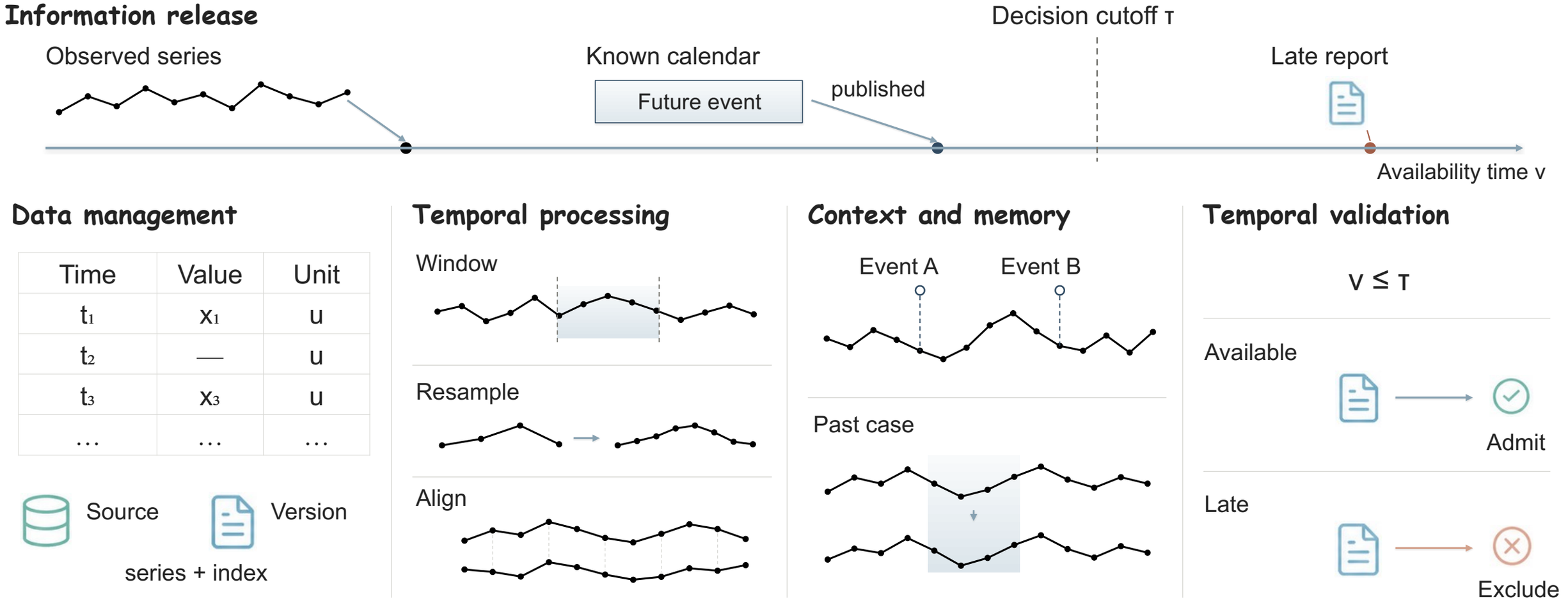

Information release
Observed series
Known calendar
Future event
published
Decision cutoff τ
Late report
Availability time v
Data management
Time
Value
Unit
t1
x1
u
t2
—
u
t3
x3
u
…
…
…
Source
Version
series + index
Temporal processing
Window
Resample
Align
Context and memory
Event A
Event B
Past case
Temporal validation
v ≤ τ
Available
Admit
Late
Exclude
LLM
Requests
Observations
Shared temporal workspace

Figure 8

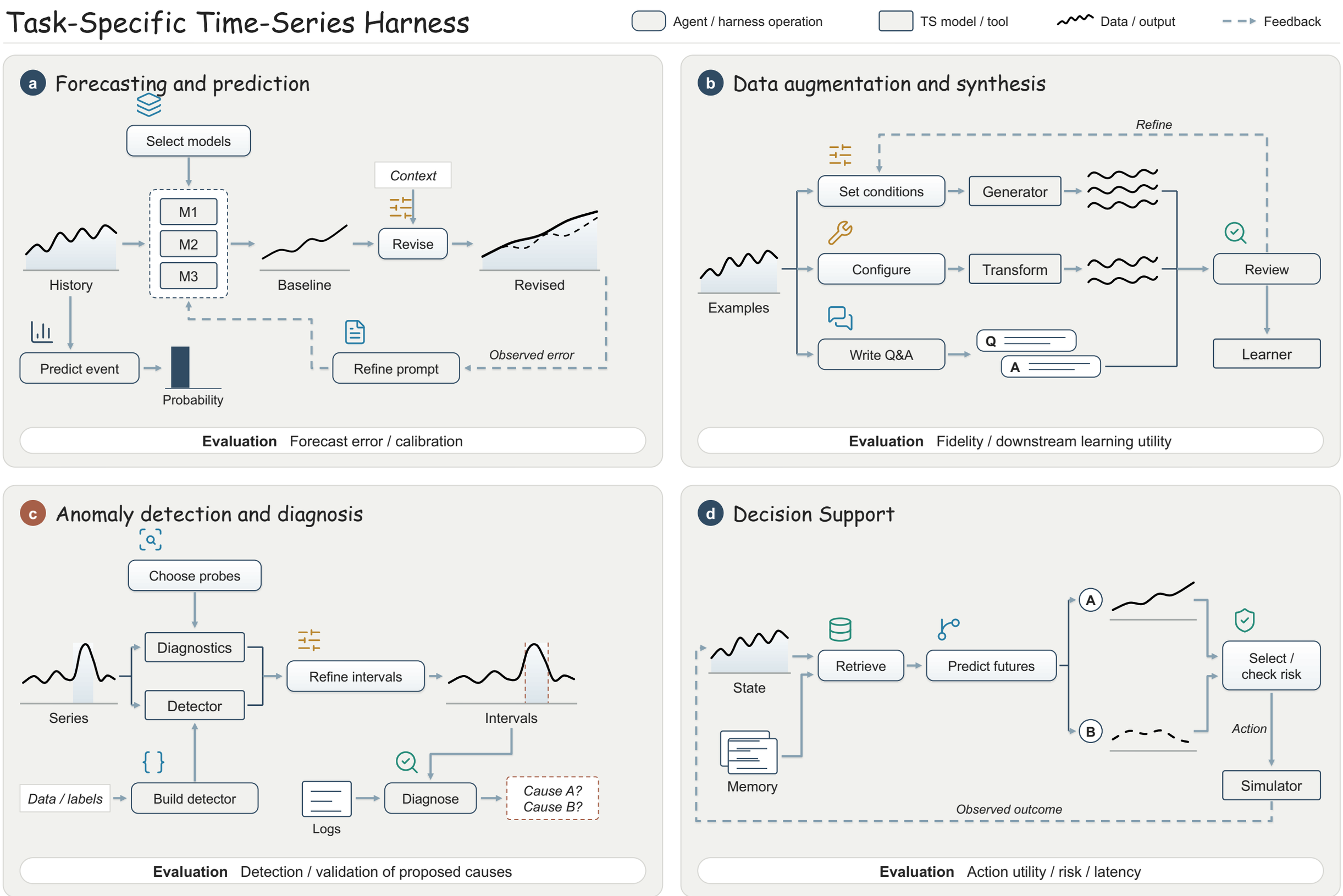

Task-Specific Time-Series Harness
Agent / harness operation
TS model / tool
Data / output
Feedback
a Forecasting and prediction
Select models
M1
M2
M3
History
Baseline
Context
Revise
Revised
Predict event
Probability
Refine prompt
Observed error
Evaluation Forecast error / calibration
b Data augmentation and synthesis
Refine
Set conditions
Generator
Configure
Transform
Examples
Review
Write Q&A
Q
A
Learner
Evaluation Fidelity / downstream learning utility
c Anomaly detection and diagnosis
Choose probes
Diagnostics
Detector
Refine intervals
Series
Intervals
Data / labels
Build detector
Logs
Diagnose
Cause A?
Cause B?
Evaluation Detection / validation of proposed causes
d Decision Support
State
Retrieve
Predict futures
A
B
Select / check risk
Action
Memory
Simulator
Observed outcome
Evaluation Action utility / risk / latency

Figure 9

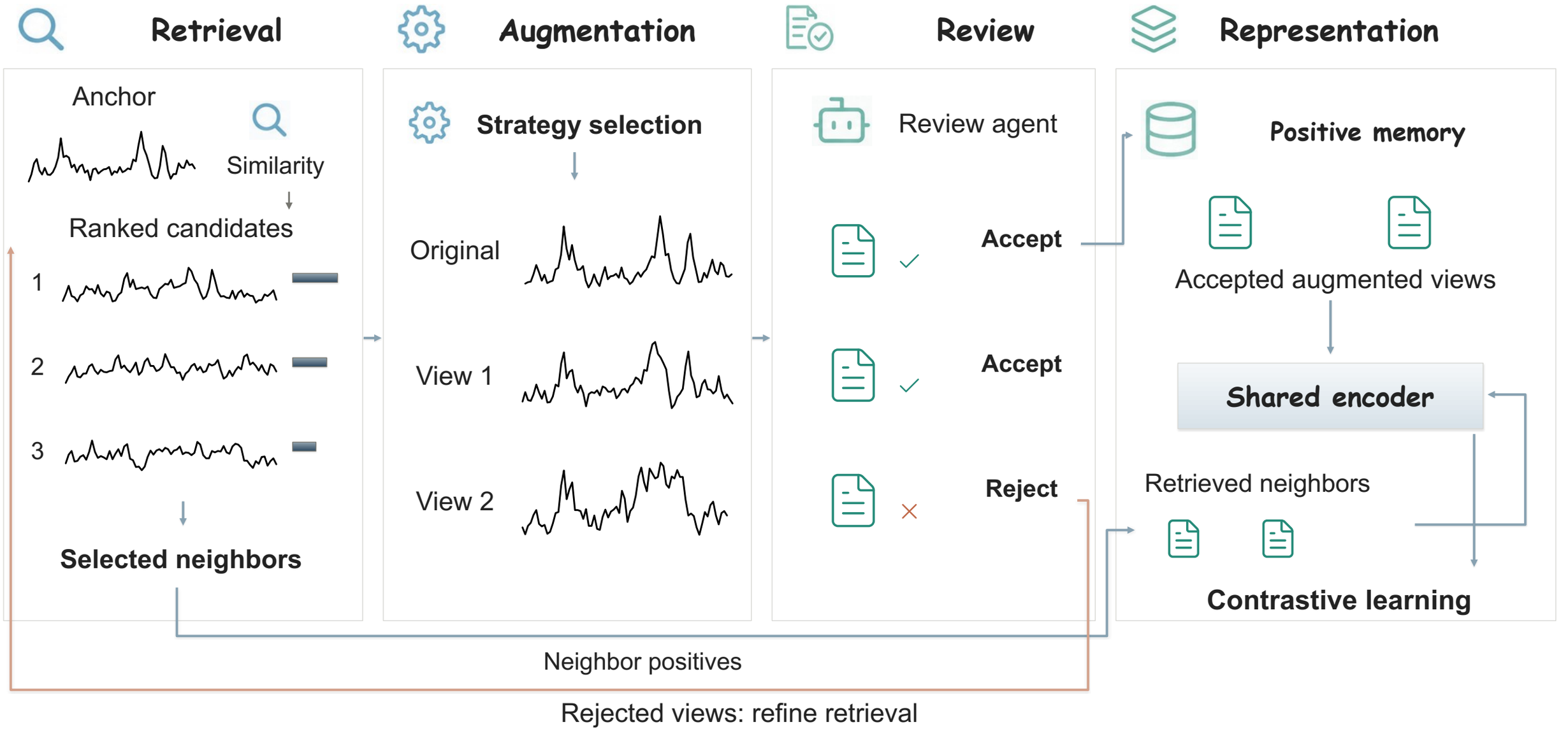

Retrieval
Augmentation
Review
Representation
Anchor
Similarity
Ranked candidates
1
2
3
Selected neighbors
Strategy selection
Original
View 1
View 2
Review agent
Accept
Accept
Reject
Positive memory
Accepted augmented views
Shared encoder
Retrieved neighbors
Contrastive learning
Neighbor positives
Rejected views: refine retrieval

Figure 10

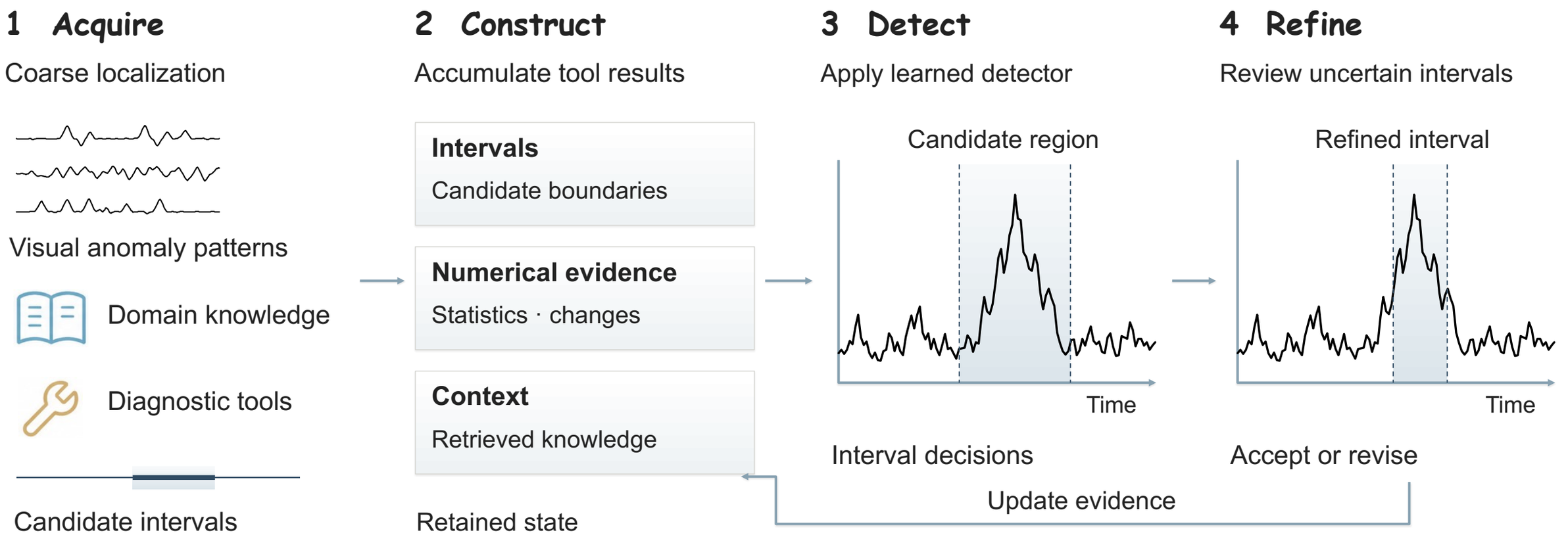

1 Acquire
Coarse localization
Visual anomaly patterns
Domain knowledge
Diagnostic tools
Candidate intervals
2 Construct
Accumulate tool results
Intervals
Candidate boundaries
Numerical evidence
Statistics · changes
Context
Retrieved knowledge
Retained state
3 Detect
Apply learned detector
Candidate region
Time
Interval decisions
4 Refine
Review uncertain intervals
Refined interval
Time
Accept or revise
Update evidence
Detector training
SFT initialization → RL
Parse validity
Valid output format
Interval agreement
False-positive penalty

Figure 11

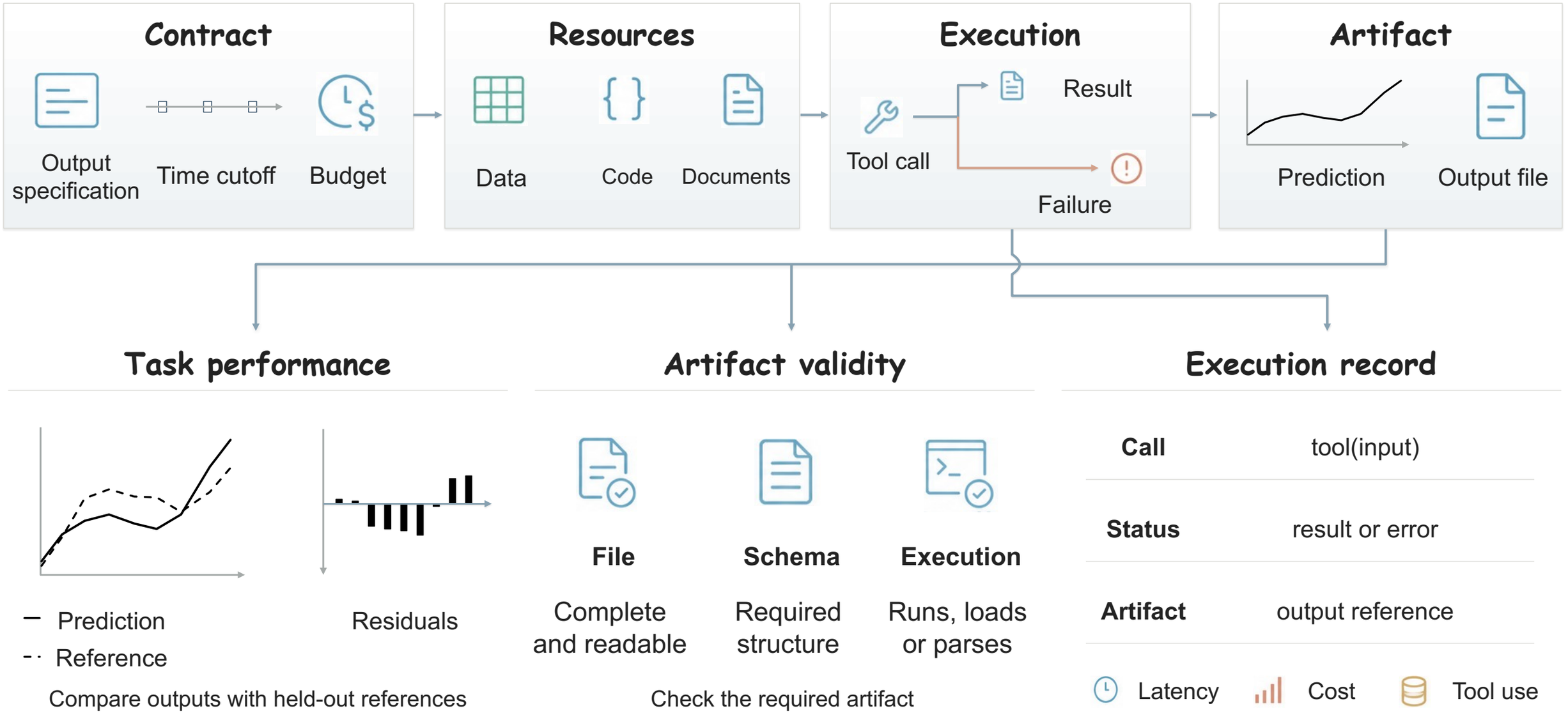

Contract
Output specification
Time cutoff
Budget
Resources
Data
Code
Documents
Execution
Tool call
Result
Failure
Artifact
Prediction
Output file
Task performance
Prediction
Reference
Residuals
Compare outputs with held-out references
Artifact validity
File
Complete and readable
Schema
Required structure
Execution
Runs, loads or parses
Check the required artifact
Execution record
Call
tool(input)
Status
result or error
Artifact
output reference
Latency
Cost
Tool use

Figure 12

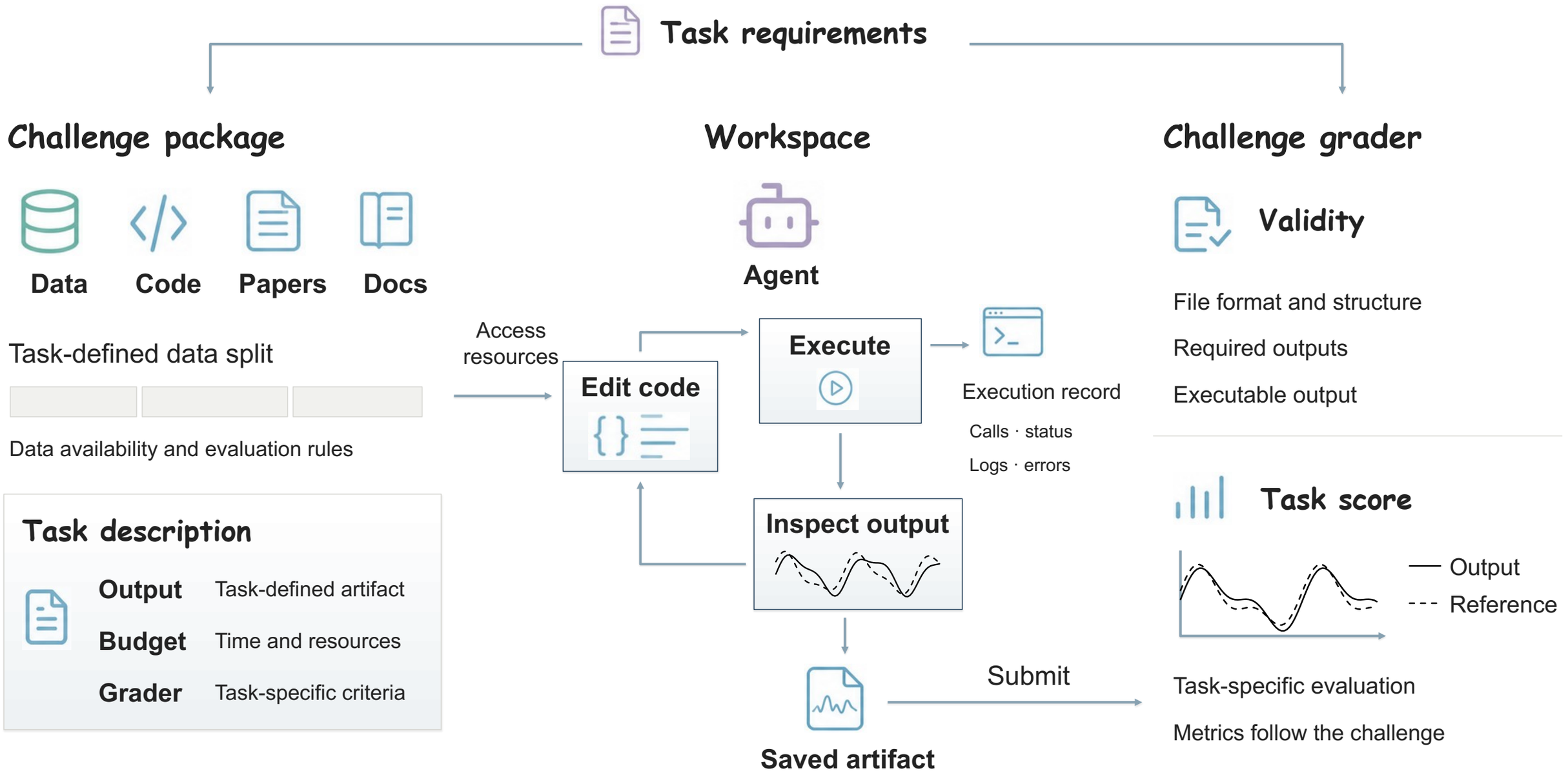
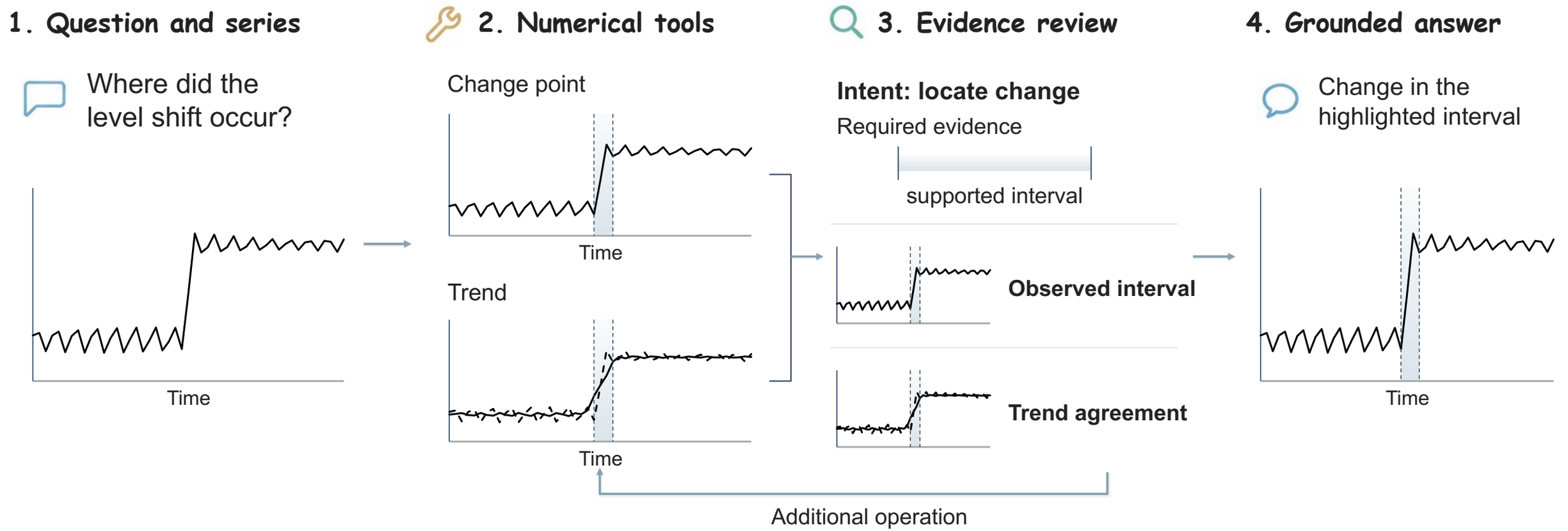

Task requirements
Challenge package
Data
Code
Papers
Docs
Task-defined data split
Data availability and evaluation rules
Task description
Output
Task-defined artifact
Budget
Time and resources
Grader
Task-specific criteria
Workspace
Agent
Access resources
Edit code
Execute
Execution record
Calls · status
Logs · errors
Inspect output
Saved artifact
Submit
Challenge grader
Validity
File format and structure
Required outputs
Executable output
Task score
Output
Reference
Task-specific evaluation
Metrics follow the challenge

Figure 13


1. Question and series
Where did the level shift occur?
Time
2. Numerical tools
Change point
Time
Trend
Time
3. Evidence review
Intent: locate change
Required evidence
supported interval
Observed interval
Trend agreement
4. Grounded answer
Change in the highlighted interval
Time
Additional operation

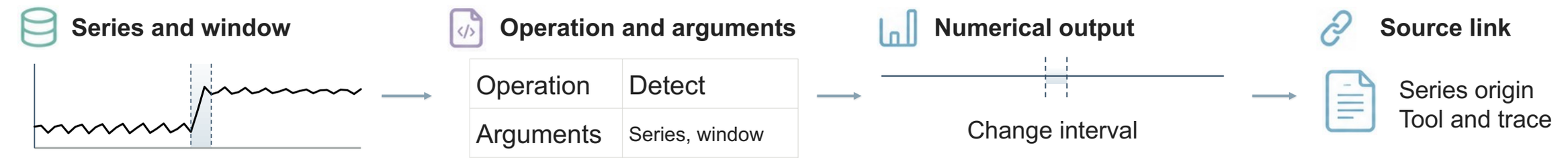

Provenance of the answer
Series and window
Operation and arguments
Operation
Detect
Arguments
Series, window
Numerical output
Change interval
Source link
Series origin
Tool and trace

Figure 14

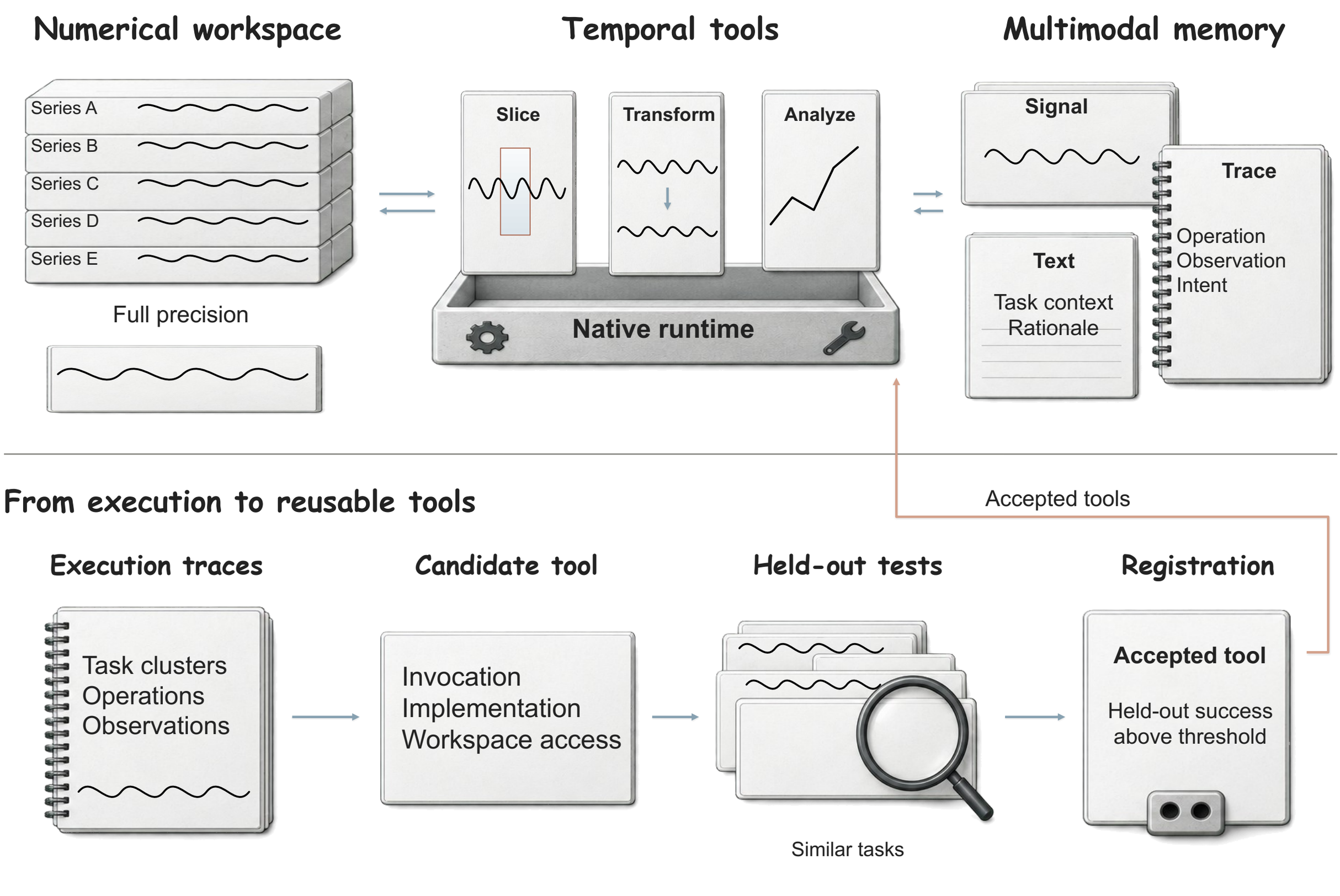
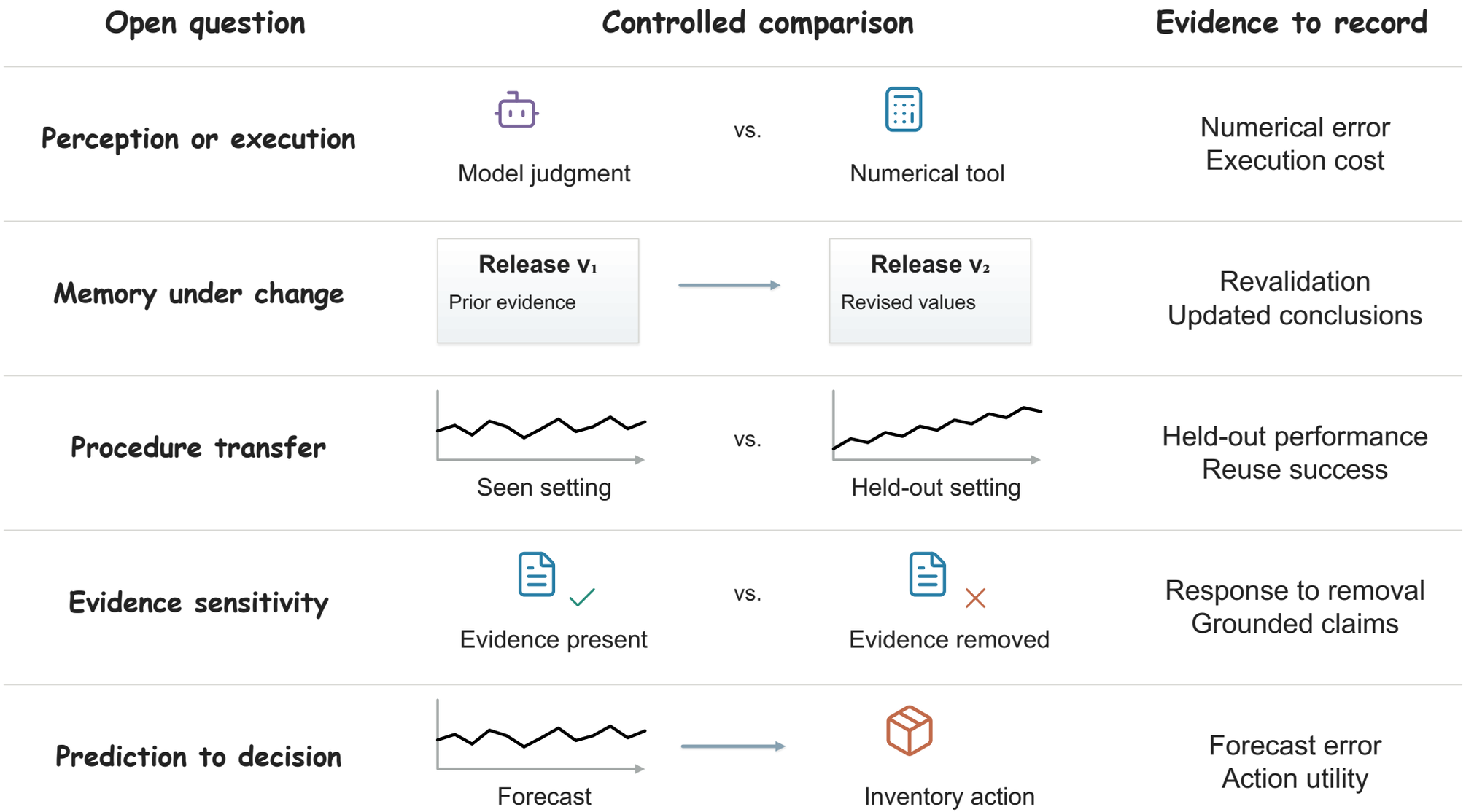

Numerical workspace
Series A
Series B
Series C
Series D
Series E
Full precision
Temporal tools
Slice
Transform
Analyze
Native runtime
Multimodal memory
Signal
Trace
Operation
Observation
Intent
Text
Task context
Rationale
From execution to reusable tools
Accepted tools
Execution traces
Task clusters
Operations
Observations
Candidate tool
Invocation
Implementation
Workspace access
Held-out tests
Similar tasks
Registration
Accepted tool
Held-out success
above threshold

Figure 15


Open question
Controlled comparison
Evidence to record
Perception or execution
Model judgment
vs.
Numerical tool
Numerical error
Execution cost
Memory under change
Release v1
Prior evidence
Release v2
Revised values
Revalidation
Updated conclusions
Procedure transfer
Seen setting
vs.
Held-out setting
Held-out performance
Reuse success
Evidence sensitivity
Evidence present
vs.
Evidence removed
Response to removal
Grounded claims
Prediction to decision
Forecast
Inventory action
Forecast error
Action utility
Match the backbone, information boundary and resource budget